An Error Budget for STAR Injection on Neutral Atoms

Counted exactly at first order, and tested blind

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The question. Transversal STAR's reach is set by one coefficient, α: a teleported rotation by θ fails with probability about $\alpha p\theta$, so simulation volume scales as $1/(\alpha p)$ (Ismail et al. 2026). The papers move every physical noise channel together with one knob, p, but a hardware team must choose which error to halve next. This note splits the injection error of QuEra's published STAR circuits by physical channel, under the paper's Table II hardware model at $p_{\mathrm{ph}} = 10^{-3}$.

The answer.

1. **Single-qubit gates carry the largest share, 43–52% at $d = 3$ to 7, but the rotation gate itself carries only 12–16%.** The rest is the Hadamards that the CNOT-to-CZ compilation leaves on data qubits. Spectators and the CZ carry about a quarter each. Initialization, measurement and atom loss (modelled, as in Table II, by a Y error) carry nothing at first order.
2. **Halving single-qubit-gate error buys ×1.32 in simulation volume at $d = 5$** (sampled: ×1.36, 95% interval 1.21–1.52). Halving the rotation gate's error alone buys ×1.08.
3. **In syndrome extraction, the CZ's correlated ZZ error, its largest term, costs this injection nothing.** Every ancilla is prepared and measured in X, so the ancilla half of a ZZ is always detected. This is specific to a single-qubit rotation gate (§4).
4. **The budget is an exact first-order count**, committed before the sampling that tested it. It predicted nine sampled quantities to within 0.95–1.10, including $d = 7$, which had not been sampled.

1. Setup

- **Circuits.** QuEra's STAR teaching circuits for the ETH QHack 2026 challenge, $d = 3$, 5, 7: $k = d$ rotated qubits, one round before the rotation; not the paper's $k = 3$ protocol. Fetched, never redistributed.
- **Noise.** Table II's channels are placed by role on the native gate set $\{R, R_X, H, \mathrm{CZ}, R_Z, M, M_X\}$ and scaled so that Eq. D1 gives $p_{\mathrm{ph}} = 10^{-3}$ (the table sums to 7.46×10^{-3}). Each CNOT becomes $H \cdot \mathrm{CZ} \cdot H$. Paired Hadamards cancel, and the rest fold into resets and measurements (Fig. 12). With every channel set equal, the rebuilt circuits reproduce the published detector error model exactly at $d = 3$, 5, 7.
- **Metric: an idealized injection error.** Noise acts only in the injection window, every detector is post-selected, later rounds are noiseless, and there is no decoder. A real protocol decodes some faults this metric rejects. Two findings forced the choice. With noisy memory rounds and a perfect unrotation, 69–84% of the challenge's reference curve's angle-dependent error is memory X errors, read out in proportion to $\sin^2\theta$. And that curve is reproduced only when matching on tsim's undecomposed error model, from which PyMatching drops every fault that flips more than two detectors (reported as tsim issue #232).

2. Method: count instead of sample

Sampling (protocol P02, 665 million shots) showed that under both the uniform and the Table II model, the injection error is one constant times $f_k(\theta)$ from $\pi/4$ to $\pi/128$, flat to ×1.15, where f_k is the misrotation term of the paper's Eq. 2. So every surviving error is a single fault that a rotation branch hides, and the constant can be counted:

- In the $\theta = 0$ Clifford proxy's error model, each fault E has a probability p_E, a set of flipped detectors D_E, and an observable bit.
- The rotation writes the patch as branches Z_S, $S \subseteq K$, with symptoms $D_S = \bigoplus_{j \in S} D_j$. A fault survives if and only if $D_E = D_S$ for some S. The pair $\{S, K \setminus S\}$ shares each symptom, and the observable bit picks the one E acts as, S^*.
- The error is $r(\theta) = \sum_E p_E \, w_{|S^*|}(\theta)$, where w_m is the closed-form failure weight of a branch pair ($w_1 = f_k$).

First-order counting of undetected faults in post-selected rotation preparation is not new:

- Choi *et al.* identify three leading-order types of undetected fault and check them against Clifford simulation (Choi et al. 2023, arXiv:2303.17380).
- Toshio *et al.* count the rotation gate's own errors and take every other single error to be detected (Toshio et al. 2024, arXiv:2408.14848).

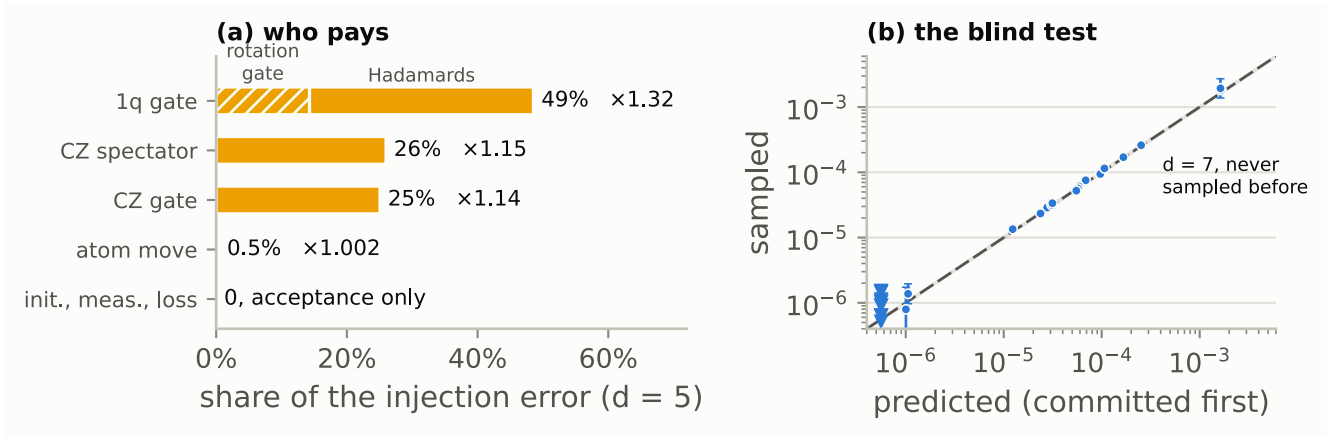


Figure 1: **(a)** Each Table II row's share of the first-order injection error at d = 5, θ = π/16 (computed); the hatched part is the rotation gate's own error. Right of each bar: what halving the row buys in simulation volume. **(b)** Protocol P04 and its CZ diagnostic: each sampled injection error (95% Wilson interval) against the prediction committed before sampling. Rows predicted to be exactly zero are drawn as 95% upper limits at the left edge.

- Liu *et al.* count error-model mechanisms that flip a logical observable without firing any detector, for Clifford injection (Liu et al. 2026, App. F).
- Veroni *et al.* describe the window around the rotation in which a phase flip goes unseen (Veroni et al. 2026). The misrotation term is the paper's Eq. 2.

What this note adds is narrow. It extends the error-model count from Clifford injection to an arbitrary angle, by matching each fault against the 2^k branch symptoms. It runs over every fault in the published circuits, channel by channel, in about 100 lines (`starlab/enumerate.py`), and it was tested blind.

3. Validation

- **Against P02 (protocol P03).** With no fitted parameter, the count reproduces P02's four sampled constants (ratios 0.97, 1.00, 1.01, 0.95) and all eight sampled modelling variants (χ^2 = 27.0 for 24 points). Only k fault classes survive, 3 of 271 at $d = 3$ and 5 of 955 at $d = 5$: each a single Z on one rotated qubit.
- **Blind (protocol P04, panel b of Figure 1).** Predictions were committed for each Table II row alone at $d = 3$ and 5, the 1q-gate row at half rate, $\theta = \pi/64$, and all of Table II at $d = 7$, before any of them was sampled (802 million shots in all). All nine scored quantities fall within 0.95–1.10 of their predictions; at $d = 7$, 2.60×10^{-4} was sampled against 2.53×10^{-4} predicted. The rows predicted at zero sampled at most 0.7% of the total (95% upper limits).
- **Where first order bends.** At full Table II strength ($s = 1$) the count is 10% low at $d = 3$ (1.10, interval 1.01–1.19). If the excess is second order it scales with p, which makes it about 1% at the operating point. At the smallest angles sampled the excess is 9–14%.

4. The budget

Table 1 gives each Table II row's share at $d = 3$, 5 and 7, and panel a of Figure 1 draws $d = 5$.

Table 1: Computed at $\theta = \pi/16$, Table II scaled to $p_{\text{ph}} = 10^{-3}$. Every surviving fault has the same weight $w_1(\theta)$, so the shares do not depend on the angle. Halving a row with share f multiplies α by $1 - f/2$: simulation volume grows by $1/(1 - f/2)$, circuit volume by its square. The 1q-gate row was also sampled halved: ×1.36 (1.21–1.52).

Table II row: share of the error (%)	d = 3	d = 5	d = 7	Halving buys (d = 5)
1q gate	43.1	48.5	51.7	×1.321
— rotation gate	11.7	14.3	15.7	×1.077
— Hadamards on data	31.3	34.3	36.0	×1.207
CZ spectator	30.5	26.0	23.4	×1.149
CZ gate	26.0	25.1	24.5	×1.143
atom move	0.5	0.5	0.4	×1.002
init., meas., both losses	0	0	0	—
A/p_{ph}	3.42	4.69	5.96	
α through Eq. 4	8.6	7.8	9.1	

Table 2: The readings the paper leaves open, all registered before any result (protocol P02). Effects are sampled, on the injection error at $\theta = \pi/16$, $d = 3$ and 5.

Choice	This note	Source	Alternative: effect
Rates	Table II × 0.134	Table II, Eq. D1	shares unchanged at first order
1q-gate channel	depolarizing	Table II (the prose says "biased", no numbers)	not priced
CNOT compilation	Hadamards cancelled and folded	Fig. 12	all kept: +20–25%
Pre-rotation feed-forward	noiseless frame update	not specified	a physical gate: +15%
CZ loss	every atom in the zone	Table II	pair only: within noise
Idle noise	none	none listed	not swept
Atom loss	a Y error	Table II	real loss is not Pauli: not priced

- **Why the rotation gate is a minority.** A Z on a rotated qubit anywhere between the rotation and the qubit's first X-check looks exactly like a branch (Choi *et al.*'s flip-projection error), and so does a Z that a stabilizer maps onto one. The Hadamards, CZs and spectators of the first syndrome round all put such Zs there. The mechanism is known; its size in these circuits is the result.
- **Initialization, measurement and loss are free at first order.** Each trips a detector that no branch trips. Loss is Table II's Y error; real atom loss is not Pauli and is not priced here. They lower acceptance, which sets the time per accepted rotation, not its error.
- **Table II gives 0.55 of the uniform model's error at every distance.**

Inside the CZ. Table II's CZ channel has three parameters: single X or Y (5×10^{-4} per term), single Z (6.3×10^{-4}), and correlated ZZ (1.3×10^{-3}).

- $\boldsymbol{ZZ}$ **costs nothing here.** A Z on an ancilla, which is prepared and measured in X, flips its measurement. That shows up in two consecutive rounds, while a branch's symptom lies in one. Planted at every window CZ, the ancilla half alone is always caught, and no ZZ has a data half that cancels it. Sampled separately and registered: 2 errors in 1.1×10^7 accepted shots at $d = 3$, none at $d = 5$.
- $\boldsymbol{X}$ **and** $\boldsymbol{Z}$ **trade places.** On a data qubit inside the Hadamard frame of an X-check, X is the term that hides; outside it, Z is; a Y never hides. An X on an ancilla also hides, by spreading through its later CZs into data Zs.
- **For hardware, with limits:** a syndrome-extraction CZ whose error is mostly ZZ beats one with the same total in single-atom terms, for this injection, to first order. This does not hold for a rotation built from two-qubit $R_{ZZ}(\theta)$ gates, where a ZZ at the rotation is the critical undetected fault (Veroni et al. 2026). In later memory rounds a ZZ is an ordinary error.

5. Limits

- **Readings.** Every choice made where the paper is silent: Table 2.
- **Scope.** QuEra's teaching circuits ($k = d$, one round before the rotation), not the paper's protocol. Distances 3–7 are sampled; 9–15 are computed by the validated method.
- $\boldsymbol{\alpha}$ **is not the paper's.** Ours is 8.6–9.1 (Table II), against the paper's ≈ 1.5 at $d = 7$ for its own protocol. The protocols differ in k, in the rounds before the rotation and in the rotation gate. Which accounts for the gap is not yet tested. Idle dephasing, which Table II omits, would add directly by the mechanism above.
- **First order** (§3). One residual is unexplained: the $d = 3$ spectator row sampled 10% above its prediction, inside tolerance.
- **Process.** Every experiment was a protocol committed before it ran, enforced by a guard. An AI agent (Claude) wrote most of the code; the companion book logs, per chapter, what the author verified by hand.

Reproduce. `github.com/mamuncseru/star-on-atoms` (book, `starlab`, protocols, ledger): `make fetch && make check`, then `python tools/exact_budget.py` and `python tools/measure_p04.py`.

Choi, Hyeongrak, Frederic T. Chong, Dirk Englund, and Yongshan Ding. 2023. *Fault Tolerant Non-Clifford State Preparation for Arbitrary Rotations.*

Ismail, Refaat, I-Chi Chen, Chen Zhao, et al. 2026. "Transversal Architecture for Megaquop-Scale Quantum Simulation with Neutral Atoms." *PRX Quantum* 7: 020343. https://doi.org/10.1103/j2fw-ccmy.

Liu, Kun, Shifan Xu, Tomas Jochym-O'Connor, Zhiyang He, Shraddha Singh, and Yongshan Ding. 2026. *In-Situ Simultaneous Magic State Injection on Arbitrary CSS qLDPC Codes.*

Toshio, Riki, Yutaro Akahoshi, Jun Fujisaki, Hirotaka Oshima, et al. 2024. *Practical Quantum Advantage on Partially Fault-Tolerant Quantum Computer.*

Veroni, Stefano, Aydin Deger, and Andrew J. Daley. 2026. *End-to-End Optimization of Partial Fault Tolerance in Neutral-Atom Arrays.*